

APPENDIX A

GENERAL NOTES

For the purpose of modelling lungs using ANSYS and simulating it, user defined function option of ANSYS is used wherein the pattern of spread of infection is fed as a code and compiled to yield the results.

A.1 BOUNDARY CONDITION

In case of Boundary condition for a linear or exponential or various disease cases, the human lung breathing pattern is fed as a sine wave as can be observed through the following code

Boundary Condition- Unsteady Mass flow rate

```
#include "udf.h"

#define PI 3.141592653589793238462643383279502884197

DEFINE_PROFILE(unsteady_massflow,thread,position)
{
    face_t f;
    real t = CURRENT_TIME;
    begin_f_loop(f,thread)
    {
        F_PROFILE(f, thread, position) = 0.000214375+0.000214375*sin(0.25*2*PI*t);
    }
    end_f_loop(f,thread)
}
```

Boundary Condition- Unsteady Pressure Outlet

```

#include "udf.h"

DEFINE_PROFILE(unsteady_pressure_outlet,thread,position)
{
face_t f;

real t = CURRENT_TIME;

begin_f_loop(f,thread)
{
F_PROFILE(f, thread, position) =-133*sin(0.25*2*PI*t);
}
end_f_loop(f,thread)
}

```

A.2 PROPERTIES OF POROUS MEDIUM

As there are limitations in ANSYS with respect to the porous medium, the porous medium model introduces an empirically derived flow resistance within a designated "porous" region of the simulation. Essentially, this model functions as an additional momentum sink within the governing momentum equations.

The porous media momentum resistance is calculated separately on each phase .

```

#include "udf.h"

#include "math.h"

#define cong 0.95

#define V17 4.72E-05

#define L17 1.39E-03

#define V18 6.86E-05

#define L18 1.17E-03

#define V19 1.02E-04

```

```
#define L19 9.89E-04
#define V20 1.56E-04
#define L20 8.34E-04
#define V21 2.45E-04
#define L21 7.04E-04
#define V22 3.93E-04
#define L22 5.94E-04
#define V23 6.48E-04
#define L23 5.01E-04
```

1. LINEAR AND EXPONENTIAL RATE OF GROWTH OF CONGESTION

NOTE: The porosity for the zones 2 and 3 can be fed through the following **udf**

```
DEFINE_PROFILE(porosityZ3,thread,position)
{
    real t = CURRENT_TIME;
    real C = 4.63e-5*t;
    face_t f;
    begin_f_loop(f,thread)
    {
        F_PROFILE(f, thread, position) = (pow(1-C,3.))/1.12;
    }
    end_f_loop(f,thread)
}
```

```
DEFINE_PROFILE(porosityZ2,thread,position)
{
    real t = CURRENT_TIME;
    real C = 4.63e-5*t;
    face_t f;
    begin_f_loop(f,thread)
    {
        F_PROFILE(f, thread, position) = (pow(1-C,2.))/1.12;
    }
    end_f_loop(f,thread)
}
```

NOTE: The permeability is fed in terms of viscous resistance for the zones 2 and 3. For linear growth of congestion pattern, the following udf is used

```

DEFINE_PROFILE(vis_res_zone3_centre_linear,thread,position)
{
    real t = CURRENT_TIME;
    real phi;
    real nu;
    real C = 4.63e-5*t;
    face_t f;
    real dna;
    real dnb1;
    real dnb2;
    begin_f_loop(f,thread)
    if (C < cong)
        {
            phi= (pow(1-C,3.))/1.12;
            nu=pow(1-phi,2.)*180;
            dna=pow(phi,3.);
            dnb2=(6*3.54e-3)/(3e8*3.14)- pow(0.000272-C*0.000272,3.);
            dnb1=pow(dnb2,2/3.);
            F_PROFILE(f, thread, position) = nu/(dna*dnb1);
        }
    else
        {
            F_PROFILE(f, thread, position) = 1.62E22;
        }
    end_f_loop(f,thread)
}

```

```

DEFINE_PROFILE(vis_res_zone2_centre_linear,thread,position)
{
    real t = CURRENT_TIME;
    real phi;
    real nu;
    real C = 4.63e-5*t;
    face_t f;
    real dna;
    real dnb171;
    real dnb172;
    real dnb181;
    real dnb182;
    real dnb191;

```

```

real dnb192;
real dnb201;
real dnb202;
real dnb211;
real dnb212;
real dnb221;
real dnb222;
real dnb231;
real dnb232;
real avgperm;
begin_f_loop(f,thread)
if (C < cong)
{
    phi= (pow(1-C,2.))/1.12;
    nu=pow(1-phi,2.)*180;
    dna=pow(phi,3.);
    dnb172=(6*V17)/(pow(2,17)*3.14)-pow(5.43E-04-C*5.43E-
04,2.)*1.5*L17;
    dnb171=pow(dnb172,2/3.);
    dnb182=(6*V18)/(pow(2,18)*3.14)-pow(5.04E-04-C*5.04E-
04,2.)*1.5*L18;
    dnb181=pow(dnb182,2/3.);
    dnb192=(6*V19)/(pow(2,19)*3.14)-pow(4.74E-04-C*4.74E-
04,2.)*1.5*L19;
    dnb191=pow(dnb192,2/3.);
    dnb202=(6*V20)/(pow(2,20)*3.14)-pow(4.51E-04-C*4.51E-
04,2.)*1.5*L20;
    dnb201=pow(dnb202,2/3.);
    dnb212=(6*V21)/(pow(2,21)*3.14)-pow(4.34E-04-C*4.34E-
04,2.)*1.5*L21;
    dnb211=pow(dnb212,2/3.);
    dnb222=(6*V22)/(pow(2,22)*3.14)-pow(4.24E-04-C*4.24E-
04,2.)*1.5*L22;
    dnb221=pow(dnb222,2/3.);
    dnb232=(6*V23)/(pow(2,23)*3.14)-pow(4.19E-04-C*4.19E-
04,2.)*1.5*L23;
    dnb231=pow(dnb232,2/3.);
    avgperm= (pow(2,17)*(dna*dnb171) + pow(2,18)*(dna*dnb181) +
pow(2,19)*(dna*dnb191) + pow(2,20)*(dna*dnb201) + pow(2,21)*(dna*dnb211)
+ pow(2,22)*(dna*dnb221) + pow(2,23)*(dna*dnb231))/(nu*(pow(2,17) +
pow(2,18) + pow(2,19) + pow(2,20) + pow(2,21) + pow(2,22) + pow(2,23)));
    F_PROFILE(f, thread, position) = 1/avgperm;
}
else
{
    F_PROFILE(f, thread, position) = 4.94E17;
}

```

```

    }
end_f_loop(f,thread)
}

```

NOTE: For exponential growth of congestion pattern, the following udf is used

```

DEFINE_PROFILE(vis_res_zone3_side1_exp,thread,position)
{
real t = CURRENT_TIME;
real phi;
real nu;
real C = 0.000000000005*pow(1+0.0011,t);
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f,thread)
if (C < cong)
{
phi= (pow(1-C,3.))/1.12;
nu=pow(1-phi,2.)*180;
dna=pow(phi,3.);
dnb2=(6*3.54e-3)/(3e8*3.14)-pow(0.000272-C*0.000272,3.);
dnb1=pow(dnb2,2/3.);
F_PROFILE(f, thread, position) = nu/(dna*dnb1);
}
else
{
F_PROFILE(f, thread, position) = 1.62E22;
}
end_f_loop(f,thread)
}

```

```

DEFINE_PROFILE(vis_res_zone2_side1_exp,thread,position)
{
real t = CURRENT_TIME;
real phi;
real nu;
real C = 0.000000000005*pow(1+0.0011,t);
face_t f;
real dna;
real dnb171;
real dnb172;
real dnb181;

```

```

real dnb182;
real dnb191;
real dnb192;
real dnb201;
real dnb202;
real dnb211;
real dnb212;
real dnb221;
real dnb222;
real dnb231;
real dnb232;
real avgperm;
begin_f_loop(f,thread)

if (C < cong)
{
    phi= (pow(1-C,2.))/1.12;
    nu=pow(1-phi,2.)*180;
    dna=pow(phi,3.);
    dnb172=(6*V17)/(pow(2,17)*3.14)-pow(5.43E-04-C*5.43E-
04,2.)*1.5*L17;
    dnb171=pow(dnb172,2/3.);
    dnb182=(6*V18)/(pow(2,18)*3.14)-pow(5.04E-04-C*5.04E-
04,2.)*1.5*L18;
    dnb181=pow(dnb182,2/3.);
    dnb192=(6*V19)/(pow(2,19)*3.14)-pow(4.74E-04-C*4.74E-
04,2.)*1.5*L19;
    dnb191=pow(dnb192,2/3.);
    dnb202=(6*V20)/(pow(2,20)*3.14)-pow(4.51E-04-C*4.51E-
04,2.)*1.5*L20;
    dnb201=pow(dnb202,2/3.);
    dnb212=(6*V21)/(pow(2,21)*3.14)-pow(4.34E-04-C*4.34E-
04,2.)*1.5*L21;
    dnb211=pow(dnb212,2/3.);
    dnb222=(6*V22)/(pow(2,22)*3.14)-pow(4.24E-04-C*4.24E-
04,2.)*1.5*L22;
    dnb221=pow(dnb222,2/3.);
    dnb232=(6*V23)/(pow(2,23)*3.14)-pow(4.19E-04-C*4.19E-
04,2.)*1.5*L23;
    dnb231=pow(dnb232,2/3.);
    avgperm= (pow(2,17)*(dna*dnb171) + pow(2,18)*(dna*dnb181) +
pow(2,19)*(dna*dnb191) + pow(2,20)*(dna*dnb201) + pow(2,21)*(dna*dnb211)
+ pow(2,22)*(dna*dnb221) + pow(2,23)*(dna*dnb231))/(nu*(pow(2,17) +
pow(2,18) + pow(2,19) + pow(2,20) + pow(2,21) + pow(2,22) + pow(2,23)));
    F_PROFILE(f, thread, position) = 1/avgperm;
}

```

```

    }
else
    {
        F_PROFILE(f, thread, position) = 4.94E17;
    }
end_f_loop(f,thread)
}

```

2. TUBERCULOSIS- The UDF is arrived at based on the case study

```

#define cong 0.975
#define t1max 200
#define t2max 500
#define t3max 3600
#define stepc 8.33e-4
DEFINE_PROFILE(first_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f, thread)
    if (t > 0 & t < t2max)
    {
        {
            real C = stepc * pow(1 + 0.0019615, t);
            F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
        }
    }
else
    if (t >= 1800 & t <= t3max)
    {
        {
            real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
            F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
        }
    }
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(first_z3_exp, thread, position)
{
    real t = CURRENT_TIME;
    real phi;
    real nu;
    face_t f;

```



```

real dna;
real dnb1;
real dnb2;
begin_f_loop(f, thread)
if (t > 0 & t < t2max)
{
real C = stepc * pow(1 + 0.0019615, t);
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
}
else
if (t >= 1800 & t <= t3max)
{
real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
}
}
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(sec_porosityZ3, thread, position)
{
real t = CURRENT_TIME;
face_t f;
begin_f_loop(f, thread)
if (t > 0 & t < t1max)
{
F_PROFILE(f, thread, position) = 0.893;
}
}

```

```

else
if (t >= t1max & t <= t2max)
{
real C = stepc * pow(1 + 0.0019615, t);
F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
}
else
if (t >= 1800 & t <= t3max)
{
real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
}
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(sec_z3_exp, thread, position)
{
real t = CURRENT_TIME;
real phi;
real nu;
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f, thread)
if (t > 0 & t < t1max)
{
F_PROFILE(f, thread, position) = 1.61E+8;
}
else
if (t >= t1max & t <= t2max)
{
real C = stepc * pow(1 + 0.0019615, t);
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
else
{

```

```

        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if (t >= 1800 & t <= t3max)
{
    real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(third_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
    face_t f;
    begin_f_loop(f, thread)
    if (t > 0 & t < 1800)
    {
        F_PROFILE(f, thread, position) = 0.893;
    }
    else
    if (t >= 1800 & t <= t3max)
    {
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(third_z3_exp, thread, position)

```

```

{
real t = CURRENT_TIME;
real phi;
real nu;
real C = stepc * pow(1 + 0.003923, (CURRENT_TIME - 1800));
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f, thread)
if (t > 0 & t < 1800)
{
F_PROFILE(f, thread, position) = 1.61E+8;
}
else
if (t >= 1800 & t <= t3max)
{
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
else
{
F_PROFILE(f, thread, position) = 1.62E22;
}
}
end_f_loop(f, thread)
}

```

3. REMEDIAL MEASURES

No Smoking

```

#define cong 0.97
#define t1max 720
#define t2max 1440
#define t3max 2160
#define t4max 2880
#define t5max 3600
#define stepc 8.33e-4

```

```

DEFINE_PROFILE(first_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f, thread)
    if(t > 0 & t <= t1max)
        {
            real C = stepc * pow(1 + 0.0096675, t);
            F_PROFILE(f, thread, position) = (pow(1 - C,3.))/1.12;
        }
    else
    if(t > t1max & t <= t5max)
        {
            F_PROFILE(f, thread, position) = 0.00303;
        }
    end_f_loop(f,thread)
}

```

```

DEFINE_PROFILE(first_z3_exp,thread,position)
{
    real t = CURRENT_TIME;
    real phi;
    real nu;
    real C = stepc * pow(1+0.0096675,t);
    face_t f;
    real dna;
    real dnb1;
    real dnb2;
    begin_f_loop(f,thread)
    if(t > 0 & t <= t1max)
        {
            if (C < cong)
                {
                    phi= (pow(1-C,3.))/1.12;
                    nu=pow(1-phi,2.)*180;
                    dna=pow(phi,3.);
                    dnb2=(6*3.54e-3)/(3e8*3.141593)-pow(0.000272-C*0.000272,3.);
                    dnb1=pow(dnb2,2/3.);
                    F_PROFILE(f, thread, position) = nu/(dna*dnb1);
                }
        }
    else
    if(t > t1max & t <= t5max)
        {
            F_PROFILE(f, thread, position) = 6.14E24;
        }
    end_f_loop(f,thread)
}

```

```

    }
end_f_loop(f,thread)
}

```

```

DEFINE_PROFILE(sec_porosityZ3,thread,position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f,thread)
    if(t > 0 & t <= t2max)
    {
        F_PROFILE(f, thread, position) = 0.893;
    }
    else
    if(t > t2max & t <= t3max)
    {
        real C = stepc * pow(1+0.008186109,(CURRENT_TIME - t2max));
        F_PROFILE(f, thread, position) = (pow(1-C,3.))/1.12;
    }
    else
    if(t > t3max & t <= t5max)
    {
        real C = stepc * pow(1+0.008186109,(720))-((CURRENT_TIME
-t3max)*0.000178586);

```

For Uncontrolled Smoking

```

    real C = stepc*pow(1+0.008186109,(CURRENT_TIME - t2max))-
((CURRENT_TIME-t3max)*0.000178586);

```

For Reduced Smoking by 90%

```

    real C = stepc*pow(1+0.008186109,(720))+
stepc*pow(1+0.00081861098*1,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);

```

For Reduced Smoking by 65%

```

    real C = stepc*pow(1+0.008186109,(720))+
stepc*pow(1+0.0008186109*3.5,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);

```

For Reduced Smoking by 50%

```

    real C = stepc*pow(1+0.008186109,(720))+

```

```

stepc*pow(1+0.0008186109*5,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);

```

```

    F_PROFILE(f, thread, position) = (pow(1-C,3.))/1.12;
}
end_f_loop(f,thread)
}

```

```

DEFINE_PROFILE(sec_z3_exp,thread,position)
{
real t = CURRENT_TIME;
real phi;
real nu;
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f,thread)
if(t > 0 & t <= t2max)
{
    F_PROFILE(f, thread, position) = 1.61E+8;
}
else
if(t > t2max & t <= t3max)
{
    real C = stepc * pow(1+0.008186109,(CURRENT_TIME - t2max));
    if (C < cong)
    {
        phi= (pow(1-C,3.))/1.12;
        nu=pow(1-phi,2.)*180;
        dna=pow(phi,3.);
        dnb2=(6*3.54e-3)/(3e8*3.141593)-pow(0.000272-C*0.000272,3.);
        dnb1=pow(dnb2,2/3.);
        F_PROFILE(f, thread, position) = nu/(dna*dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if(t > t3max & t <= t5max)
{
    real C = stepc * pow(1+0.008186109,720)- ((CURRENT_TIME- t3max)*

```

0.000178586);

For Uncontrolled Smoking

```
real C = stepc*pow(1+0.008186109,(CURRENT_TIME - t2max))-
((CURRENT_TIME-t3max)*0.000178586);
```

For Reduced Smoking by 90%

```
real C = stepc*pow(1+0.008186109,(720))+
stepc*pow(1+0.00081861098*1,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);
```

For Reduced Smoking by 65%

```
real C = stepc*pow(1+0.008186109,(720))+
stepc*pow(1+0.0008186109*3.5,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);
```

For Reduced Smoking by 50%

```
real C = stepc*pow(1+0.008186109,(720))+
stepc*pow(1+0.0008186109*5,(CURRENT_TIME - t3max))-
((CURRENT_TIME-t3max)*0.000178586);
```

```
if (C < cong)
{
phi= (pow(1-C,3.))/1.12;
nu=pow(1-phi,2.)*180;
dna=pow(phi,3.);
dnb2=(6*3.54e-3)/(3e8*3.141593)-pow(0.000272-C*0.000272,3.);
dnb1=pow(dnb2,2/3.);
F_PROFILE(f, thread, position) = nu/(dna*dnb1);
}
else
{
F_PROFILE(f, thread, position) = 9.416E23;
}
}
end_f_loop(f,thread)
}
```

4. REMEDIAL MEASURES FOR TUBERCULOSIS

Normal Immunity


```

#define cong 0.975
#define t1max 200
#define t2max 500
#define t3max 1440
#define t4max 2520
#define t5max 2640
#define t6max 3060
#define stepc 8.33e-4
DEFINE_PROFILE(first_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f, thread)
    if (t > 0 & t <= t2max)
    {
        real C = stepc * pow(1 + 0.0019615, t);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t3max & t <= t4max)
    {
        real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t4max & t <= t5max)
    {
        real C = stepc * pow(1 + 0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t5max & t <= t6max)
    {
        real C = stepc * pow(1 + 0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
}

```

```

end_f_loop(f, thread)
}

DEFINE_PROFILE(first_z3_exp, thread, position)
{
real t = CURRENT_TIME;
real phi;
real nu;
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f, thread)
if (t > 0 & t <= t2max)
{
real C = stepc * pow(1 + 0.0019615, t);
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
}
else
if (t > t3max & t <= t4max)
{
real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
For Poor Immunity
real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
else
{

```

```

        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if (t > t4max & t <= t5max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if (t > t5max & t <= t6max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {

```

```

        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(sec_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f, thread)
    if (t > 0 & t < t1max)
    {
        F_PROFILE(f, thread, position) = 0.893;
    }
    else
    if (t >= t1max & t <= t2max)
    {
        real C = stepc * pow(1 + 0.0019615, t);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t3max & t <= t4max)
    {
        real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t4max & t <= t5max)
    {
        real C = stepc * pow(1 + 0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t5max & t <= t6max)
    {
        real C = stepc * pow(1 + 0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
        For Poor Immunity

```

```

        real C = stepc * pow(1 + 0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
end_f_loop(f, thread) }

```

```

DEFINE_PROFILE(sec_z3_exp, thread, position)
{
    real t = CURRENT_TIME;
    real phi;
    real nu;
    face_t f;
    real dna;
    real dnb1;
    real dnb2;
    begin_f_loop(f, thread)
    if (t > 0 & t < t1max)
    {
        F_PROFILE(f, thread, position) = 1.61E+8;
    }
    else
    if (t >= t1max & t <= t2max)
    {
        real C = stepc * pow(1 + 0.0019615, t);
        if (C < cong)
        {
            phi = (pow(1 - C, 3.)) / 1.12;
            nu = pow(1 - phi, 2.) * 180;
            dna = pow(phi, 3.);
            dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
            dnb1 = pow(dnb2, 2 / 3.);
            F_PROFILE(f, thread, position) = nu / (dna * dnb1);
        }
        else
        {
            F_PROFILE(f, thread, position) = 1.62E22;
        }
    }
    else
    if (t > t3max & t <= t4max)
    {
        real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
For Poor Immunity
        real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
    }
}

```

```

    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}

else
if (t > t4max & t <= t5max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}

else
if (t > t5max & t <= t6max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);

```

```

        if (C < cong)
        {
            phi = (pow(1 - C, 3.)) / 1.12;
            nu = pow(1 - phi, 2.) * 180;
            dna = pow(phi, 3.);
            دنب2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
            دنب1 = pow(دنب2, 2 / 3.);
            F_PROFILE(f, thread, position) = nu / (dna * دنب1);
        }
        else
        {
            F_PROFILE(f, thread, position) = 1.62E22;
        }
    }
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(third_porosityZ3, thread, position)
{
    real t = CURRENT_TIME;
    face_t f;
    begin_f_loop(f, thread)
    if (t > 0 & t <= t3max)
    {
        F_PROFILE(f, thread, position) = 0.893;
    }
    else
    if (t > t3max & t <= t4max)
    {
        real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
    else
    if (t > t4max & t <= t5max)
    {
        real C = stepc * pow(1 + 0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
        For Poor Immunity
        real C = stepc * pow(1 + 0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
        F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
    }
}

```

```

else
if (t > t5max & t <= t6max)
{
real C = stepc * pow(1 + 0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
For Poor Immunity
real C = stepc * pow(1 + 0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);
F_PROFILE(f, thread, position) = (pow(1 - C, 3.)) / 1.12;
}
end_f_loop(f, thread)
}

```

```

DEFINE_PROFILE(third_z3_exp, thread, position)
{
real t = CURRENT_TIME;
real phi;
real nu;
face_t f;
real dna;
real dnb1;
real dnb2;
begin_f_loop(f, thread)
if (t > 0 & t <= t3max)
{
F_PROFILE(f, thread, position) = 1.61E+8;
}
else
if (t > t3max & t <= t4max)
{
real C = stepc * pow(1 + 0.005450, (CURRENT_TIME - t3max));
For Poor Immunity
real C = stepc * pow(1 + 0.0029104, (CURRENT_TIME - t2max));
if (C < cong)
{
phi = (pow(1 - C, 3.)) / 1.12;
nu = pow(1 - phi, 2.) * 180;
dna = pow(phi, 3.);
dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
dnb1 = pow(dnb2, 2 / 3.);
F_PROFILE(f, thread, position) = nu / (dna * dnb1);
}
else
{

```



```

        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if (t > t4max & t <= t5max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-((CURRENT_TIME-t4max)*4*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-((CURRENT_TIME-t4max)*4*0.0001462);
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {
        F_PROFILE(f, thread, position) = 1.62E22;
    }
}
else
if (t > t5max & t <= t6max)
{
    real    C    =    stepc    *    pow(1    +    0.005450,
(1080))-4*0.0002237*120-((CURRENT_TIME-t5max)*2*0.0002237);
    For Poor Immunity
    real    C    =    stepc    *    pow(1    +    0.0029104,
(2020))-4*0.0001462*180-((CURRENT_TIME-t5max)*2*0.0001462);
    if (C < cong)
    {
        phi = (pow(1 - C, 3.)) / 1.12;
        nu = pow(1 - phi, 2.) * 180;
        dna = pow(phi, 3.);
        dnb2 = (6 * 3.54e-3) / (3e8 * 3.141593) - pow(0.000272 - C *
0.000272, 3.);
        dnb1 = pow(dnb2, 2 / 3.);
        F_PROFILE(f, thread, position) = nu / (dna * dnb1);
    }
    else
    {

```

```
        F_PROFILE(f, thread, position) = 1.62E22;  
    }  
}  
end_f_loop(f, thread)  
}
```